

Clinical Consultation Note

Patient ID: 8842-BC

Date of Examination: March 26, 2026

Provider: Dr. Aris Thorne

Clinical Indication: Patient presented with a firm, non-mobile 2.4cm mass in the upper outer quadrant of the left breast. No significant axillary lymphadenopathy noted on palpation.

Diagnostic Imaging & Morphological Analysis

The following metrics were captured via digital image processing of the biopsy sample to evaluate nuclear features:

Metric	Value	Units/Scale
Radius (mean)	17.99	mm (distance from centre to points on the perimeter)
Texture (mean)	10.38	Gray-scale value (standard deviation)
Perimeter (mean)	122.80	mm
Area (mean)	1001.00	mm^2
Smoothness (mean)	0.1184	Local variation in radius lengths
Compactness (mean)	0.2776	$\frac{\text{perimeter}^2}{\text{area}} - 1.0$
Concavity (mean)	0.3001	Severity of concave portions of the contour
Concave Points (mean)	0.1471	Number of concave portions of the contour

Observations

The nuclear morphology shows significant **pleomorphism**. The elevated **concavity** and **concave points** suggest irregular nuclear membranes, which are common indicators of malignancy. The **area_mean** of 1001.0 indicates a significantly enlarged nuclear volume compared to typical healthy ductal cells.

Plan

- Correlate these morphological metrics with the full pathology report.
- Schedule a follow-up PET/CT scan to assess for systemic involvement.
- Referral to Surgical Oncology for a definitive treatment plan.